Sanjay Kumar Bairi

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SUMMARY

Data Professional with 2+ years architecting end-to-end analytics solutions across data engineering, analysis, and machine learning. AWS Certified Data Engineer delivering $172K+ in measurable ROI through predictive models (84-89% accuracy), optimized ETL pipelines, and strategic dashboards. Proven ability to transform complex data into actionable insights that drive revenue growth and operational efficiency.

SKILLS

Programming & Analytics:	Python (Pandas, Numpy, Matplotlib, Seaborn), SQL, R, Java
Data Analysis :	EDA, Statistical analysis, trend analysis, data cleaning, feature engineering.
Data Visualization:	Power BI, Tableau, Excel, Matplotlib, Seaborn.
Databases:	PostgreSQL, SQL Server, Snowflake, MySQL.
Big Data & Cloud :	Databricks, PySpark, Apache Spark, AWS (S3, Glue, Lambda, Redshift, Athena, EC2), Azure (Azure SQL Database, Azure Data Factory, Azure Synapse)
ETL/ Data Pipelines:	Airflow, dbt, ETL/ELT workflow, ADF pipelines
Tools :	Git, Jupyter Notebook, VS Code, Jira, Confluence, Google Collab

EXPERIENCE

Data Analyst | Smart internz — **Jan 2022-2023 | India**

- Designed analytical data models using Star and Snowflake Schemas, improving reporting query performance.
- Developed data transformation workflows with Python and SQL, reducing manual data preparation effort.
- Built 15+ interactive financial dashboards in Power BI and Tableau with drill-down analytics, dynamic filters, and KPI scorecards, increasing stakeholder visibility into key business metrics and improving decision-making efficiency by 40%
- Executed rigorous data validation and quality assurance processes using SQL and Python, maintaining 99% accuracy in financial reporting and ensuring compliance with data
- Collaborated with cross-functional stakeholders to elicit requirements, perform gap analysis, and deliver actionable insights through ad-hoc analysis, trend reporting, and predictive analytics that supported strategic business initiatives.
- Implemented advanced SQL performance tuning techniques (indexing, query optimization, execution plan analysis) reducing average query runtime by 45% and improving dashboard refresh times from 8 minutes to 2 minutes for 50+ concurrent users.
- Conducted trend analysis and forecasting using Python (Pandas, NumPy, Matplotlib) and statistical methods, identifying seasonal patterns and business cycles that informed inventory planning, resulting in $180K cost savings through optimized stock levels.

Data Scientist | IBM — **Jan 2021-2022 | India**

- Engineered end-to-end machine learning pipelines processing records using Python (Pandas, NumPy, Scikit-learn, Matplotlib) and SQL, implementing feature engineering and data preprocessing techniques that improved model input quality by 25% and reduced data preparation time from 3 days to 8 hours
- Built and deployed 8+ predictive models using supervised learning algorithms (Logistic Regression, Random Forest, XGBoost, SVM) achieving 85%+ accuracy for classification tasks including customer churn prediction, fraud detection, and demand forecasting –
- Developed interactive ML model dashboards and performance monitoring tools in Power BI and Tableau, visualizing model metrics (accuracy, precision, recall, AUC-ROC), feature importance, and prediction distributions to communicate results to non-technical stakeholders –
- Performed advanced exploratory data analysis (EDA) and statistical modeling using hypothesis testing, A/B testing frameworks, correlation analysis, and multivariate regression to identify significant predictors and validate model assumptions
- Deployed ML models to production environments using Python APIs and cloud-based inference services, establishing automated retraining pipelines that refreshed predictions monthly, maintaining 87%+ model accuracy across 12-month production lifecycle.
- Implemented model validation and cross-validation strategies (stratified k-fold, holdout testing) ensuring robust generalization and preventing overfitting, with results documented in technical reports reviewed by senior data science leadership.

Student Technical Intern – Vaagdevi Engineering College — **August 2020- 2020 | India**

- Developed and maintained full-stack applications for university projects using HTML, CSS, JavaScript, and Python/Node
- Implemented UI components, backend routes, and basic data handling to support project functionality.
- Collaborated with faculty teams to troubleshoot issues, optimize performance, and deliver working prototypes on time.

College Club

- Built a fully functional Campus Event Management App in 24 hours, enabling student registration, event scheduling, and real-time updates; developed the full stack using HTML/CSS/JS, Python/Node, and a lightweight database.

EDUCATION

Master's in Information Systems | Auburn University at Montgomery, USA | GPA: 3.7 | **Jan 2024- Dec 2025**

Bachelors, Computer Science | Vaagdevi Engineering College, India |GPA 3.3 | **Aug 2019- Jul 2023**

Projects

E-Commerce Business Intelligence & Customer Analytics GitHub **|** Portfolio

- Analyzed 391K+ transactions, 4.2K customers, 3.6K products, generating $6.82M revenue.
- Conducted RFM segmentation, identifying Champions (940), Loyal Customers (983), At-Risk (766), and calculated $1.51M total CLV**.**
- Built Random Forest churn model (89% accuracy) and performed market basket analysis to guide cross-selling strategies.
- Developed interactive dashboards and revenue forecasts predicting $1.16M in 30-day revenue (+74% growth) to support strategic decisions.

Customer Churn Prediction System | Machine Learning **GitHub** **|** **Portfolio**

Python, Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn

Built end-to-end ML classification system analyzing 7K+ telecom customer records to predict churn risk. Performed EDA, feature engineering, and model comparison (Logistic Regression, Random Forest, XGBoost). Created risk segmentation framework and delivered business recommendations with ROI analysis.

- Engineered ML churn prediction system using Logistic Regression, Random Forest, and XGBoost achieving 84% ROC-AUC; identified 87 high-risk customers and projected $21.8K annual savings with 251% ROI through data-driven retention strategies.

Retail Sales Analysis System | SQL & Business Intelligence **GitHub** **|** **Portfolio**

SQL (SQLite), Python, Pandas, Matplotlib, Seaborn

Built SQL-focused analytics platform analyzing 9,994 retail orders using advanced querying (window functions, CTEs, complex joins). Performed RFM customer segmentation, profitability analysis, and seasonal trend identification. Delivered actionable recommendations with quantified financial impact.

- Developed SQL analytics system with 60+ advanced queries (window functions, CTEs, joins) analyzing 9,994 orders; performed RFM segmentation identifying top 10% customers generating 50%+ of $2.3M revenue and delivered targeted retention strategies worth $150K annually

Real-Time Weather Data Pipeline & Notification System Portfolio **| Dec 2024**

Technologies: Python, Apache Airflow, AWS (S3, Lambda, SES, IAM), OpenWeather API

- Deployed automated ETL pipeline extracting weather forecasts from OpenWeather API, reducing data collection.
- Built event-driven architecture using AWS Lambda triggers on S3 bucket writes, enabling instant user notifications when weather thresholds exceeded
- Orchestrated scheduled data ingestion using Apache Airflow DAGs, ensuring 99.5% uptime and consistent data availability for downstream analysis
- Applied AWS security best practices with IAM roles for least-privilege access, preparing data infrastructure for enterprise-scale analytics

CERTIFICATIONS

Google Advanced Data Analytics	Certificate
Microsoft AI & ML Engineering	Certificate
AWS Certified Cloud Practitioner	Certificate
AWS Certified Data Engineer	Certificate
Databricks Cloud Integration Certification	Certificate
Databricks Lakehouse Fundamentals	Certificate